INNOPOLIS UNIVERSITY

SNE MASTER TRACK

COURSE: [F25] Computer Systems and Networks

CSN Lab 1 Assignment

Identification of a System

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1. What is a System

System: A system is a structure made up of various components, where each component serves a specific purpose or performs a function, and collectively provides an output result. According to modern systems theory, the main components of a system are input, output, processing, and feedback **[1], [2].**

Examples-

- **Computer System:**
    - **Input:** This part receives data from the keyboard, mouse or other input devices, ports and sends it to the processing unit.
    - **Processing:** The main function of this part is to process the information received from the input using the CPU, memory, GPU, and RAM.
    - **Output:** Displays the results through a monitor, printer, or external output device.
    - **Feedback:** The user issues the next command after seeing the output **[3], [4].** Also, the system compares the input with the output to find out errors and send feedback to adjust or fix the process.
- **Gaming Consoles (Such as Nintendo 16-bit):**
    - **Input Unit:** Controller, joystick, games card, etc.
    - **Processing Unit:** This unit uses the console's processor and graphics chip to create the game's logic and video graphics.
    - **Output Unit:** Visual images displayed on a TV or monitor, sound on speakers.
    - **Feedback Unit:** This part changes the state of the game according to the player's response **[3], [4].**

2. Basic Components and Diagram of an Effective System

The main components of an effective system are:

1. **Input:** Information or instructions that are provided to the system.
2. **Processing:** The main function of this component is to process the information received from the input and produce results.
3. **Output:** Providing processed results for the client, acting as a bridge for communication, and preparing the data for the user to understand.
4. **Feedback:** Monitoring the output to provide information to adjust the system, providing the system with information to find and resolve errors, refining the performance of the system through output monitoring, and signaling messages or notifications to the user.

Diagram:

[**INPUT** (keyboard, LAN, Ethernet, Mouse, Slots] → [**PROCESSING** (CPU, GPU, Hard Drive, RAM] → [**OUTPUT** (Display)]

↑

[**FEEDBACK** (Beep Sounds, Notifications)]

Fig. 1: Diagram of an Effective System

- Input → Processing → Output
- Output → Feedback → Processing Adjustment

3. Feedback on the System and Its Use?

Feedback on the System: Feedback is information, corrections, or warnings about the output that are fed back to the system and used to control its activities to produce the desired output, signal the user to take appropriate action, and help make adjustments. This creates a closed-loop, allowing the system to automatically adjust.

Uses of Feedback:

- Used to improve performance.
- Used to detect errors or deviations.
- Provides signals to the user.
- Used to maintain system stability **[3], [4].**

4. Information System and It's Example

Information Systems: An information system is an integrated system that collects, stores, processes, and distributes information for decision-making, control, and coordination **[5], [6], [7].** People, hardware, software, data, and networks work together in this integrated system.

Examples:

- **Bank MIS:** Helps management by generating deposit, withdrawal and transaction reports **[8].** Along with this, transaction devices exchange, provide and control customer information. ATMs receive, store and control transaction data during online transactions.
- **Social Media (Such as Twitter or Instagram):** Information systems play an important role for social media. Here, billions of people are connected to each other like a web. Where there are-
    - **Hardware:** Data centers, servers
    - **Software:** Apps, web interfaces
    - **Data:** Users, posts and information
    - **People:** Users, IT staff
    - **Process:** Data exchange, analysis **[7]**

5. Components and Diagrams of an Effective Information System

In an information system, many types of components work for data reception, process, manipulation, visualization. In this system, people are connected to each other, to data and to themselves, as well as devices are connected to each other. An effective information system is

the banking system and its information system, which is the information system of internet banking system. The main components of this important system are:

- User
- Accessing Software & Hardware
- Internet & Security Protocols
- Servers
- Processing
- Database
- Response
- System Management **[5], [6], [9]–[11]**

Diagram Instructions:

- Requests and responses are sent to the application or hardware through the network (Internet).
- Data (Travel Requests) is recorded and routed by encryption and accessed through the network to send data to web application servers.
- After back-end processing or user authentication, the user's request, like doing a transaction or checking a balance, is run through the application web server.
- Data are retrieved or updated by the application server from the database using a **DBMS** (Database Management Server).
- Requested information is sent back to the application web server and is strictly monitored by IT administrators (bank staff) for monitoring to ensure the entire processing runs smoothly.
- Finally, Clients or users can easily find their requests and the responses to do secure banking.

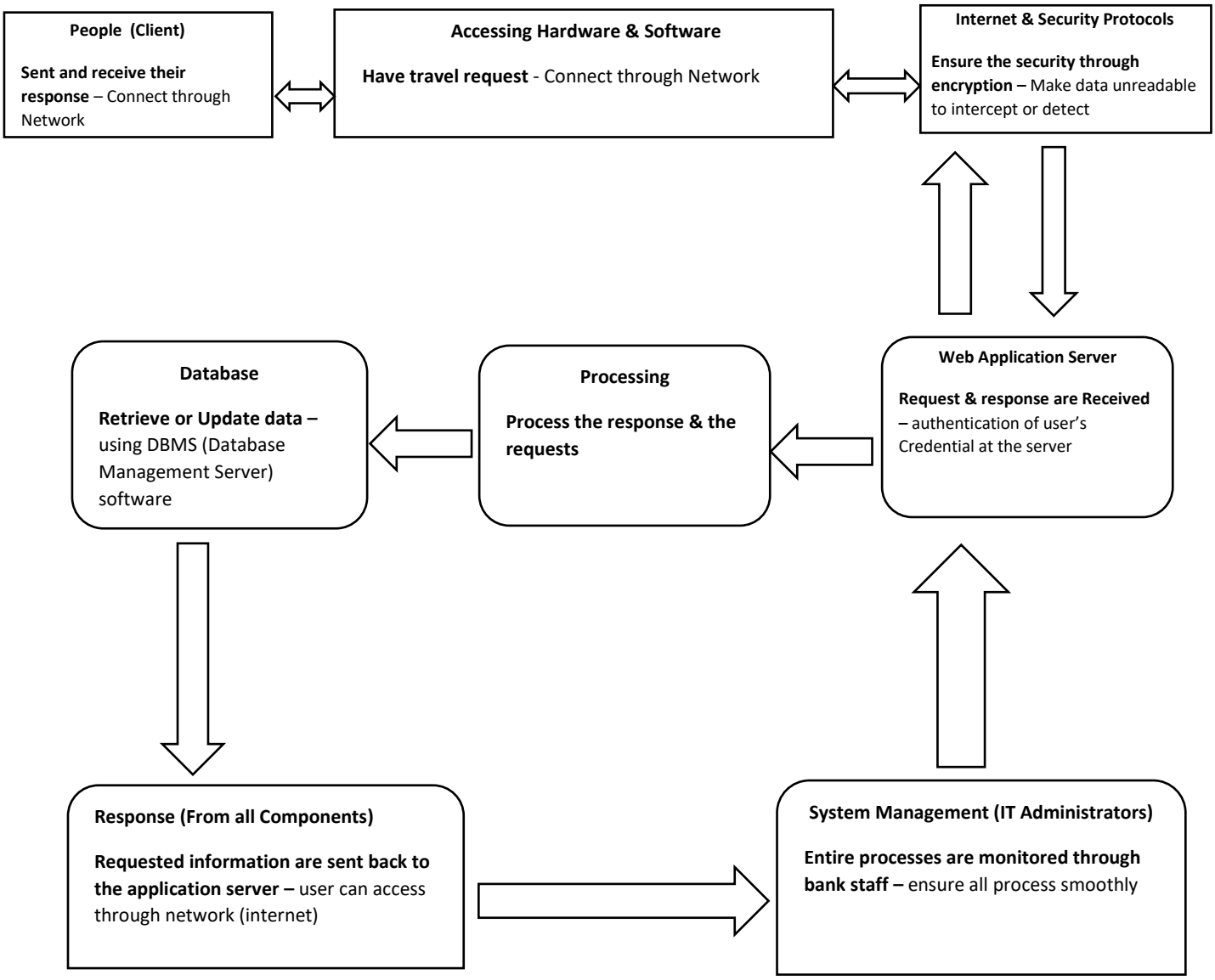


Fig. 2: Working Information System of Online Banking.

6. Components of a Distributed File System

A distributed file system is a special type of system that distributes the task of storing and managing files across multiple computers or servers. To the user, it looks like a single file system,

but the data inside is spread across multiple locations. Its main purpose is to increase data access speed, tolerate system failures, and ensure availability for the user.

Key Components:

1. **Node:** Computers or servers that store and process data. Each node can operate independently.
2. **File Server:** The main file keeper and the center for distributing data to the nodes.
3. **File Client:** The user's means of accessing files, which connects to the server.
4. **Network:** The medium for exchanging data between nodes and servers.
5. **Namespace Manager:** Controls the name, location, and version of files, so that the user can access data without directly knowing the location of the node.
6. **Cache:** Speeds up file access by storing data locally.

Main Functions:

- This type of file system primarily stores and distributes files.
- It creates and controls replication, which properly locates, stores, and removes files at specific locations.
- It also provides a unique user experience **[12], [13], [14]**.

Diagram Instructions:

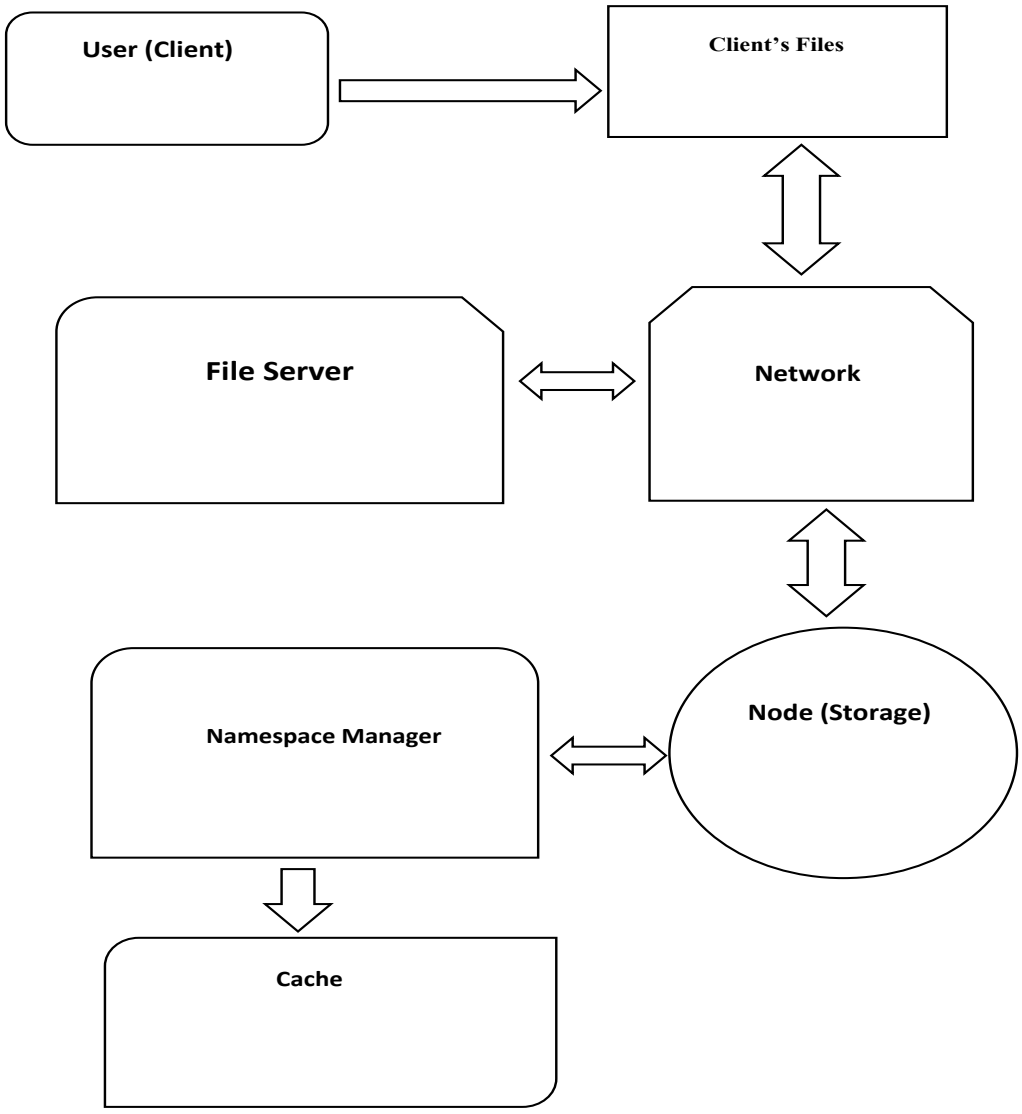


Fig. 3: Distributed File System

7. Differences Between Centralized, Decentralized and Distributed Systems

Centralized systems have a single point of control, while decentralized systems have some independence, and distributed systems make the entire system transparent and tolerant to the user. The differences between them in **table-1** are as follows:

Table-1

System type	Definition	Features	Examples
Centralized	All data and processing takes place on a single server.	Single control - maintenance is easy and the entire system stops if the server fails	Bank's main server, University file server, Google Drive **[12]**
Decentralized	There are multiple servers, but they work independently.	Partial control - Some servers fail while others remain operational, which can be a bit more complicated to manage.	Servers in branch offices in various cities, BitTorrent **[13]**
Distributed	Information and processing are divided into multiple nodes, making it appear to the user as a single system.	Load is evenly distributed - failure tolerant and remains transparent to users.	Google File System, Amazon S3, Amazon EFS **[14], [15]**

8. Transparency and Examples in Distributed Systems

Transparency in Distributed Systems: Transparency is a fundamental feature of distributed systems, which makes them easier to use and more reliable. It hides the complex communication processes and allows users and developers to focus on solving the main problems. Various transparencies, including access, replication, location, concurrency, failure, play an important role in building advanced and effective distributed systems.

Types of Transparency with Examples:

1. **Location Transparency:** The basic idea of location transparency is that the user does not need to know in which country, on which server, or in which physical location the data or resource is actually located. When the user accesses the data, the system acts as if

everything is in one place. As a result, the user experience remains unchanged even if the data is moved from one server to another.

Example: DNS hides the location of the server by converting the domain name to the IP address. Similarly, a virtual machine also hides the location of the actual hardware.

2. **Access Transparency:** It allows the user and the application to access the resource whether it is on the local machine or on a remote server, the method is exactly the same. This makes it much easier for the developer to use the system because there is no need to learn different commands, which also saves time.

   Example: Using RPC (Remote Procedure Call), the functions of the remote server can be used like local functions, whereas the message queue system hides the complexity of communication from the user.

3. **Concurrency Transparency:** There can be problems when many users or processes want to use a shared resource at the same time. Concurrency transparency hides these problems and ensures that data will not be corrupted or inconsistent even if multiple users work together.

   Examples: Transaction management (ACID property) and locking systems keep data correct and secure.

4. **Replication Transparency:** In distributed systems, multiple copies of the same data are often kept on separate servers for reliability and fast access. Replication transparency ensures that the user will never realize that the data he is using is actually stored in copies on many servers.

   Examples: CDN (Content Delivery Network) and database replication systems.

5. **Failure Transparency:** Even if any part of the system is damaged or a server is shut down, the user's service should not be interrupted—this is failure transparency. It keeps the system running continuously and solves the problem by using backup or alternative servers.

   Example: Load balancer automatically bypasses the failed server and sends the request to another healthy server.

6. **Security Transparency:** The user does not have to worry about separate security measures, but his data remains encrypted and protected. As a result, the security of the system is ensured without the user's knowledge.

   Example: Encryption, access control mechanism.

7. **Management Transparency:** As distributed systems grow, it becomes difficult to control and monitor them. Using management transparency, the system can be easily controlled from a single interface and using it, the admin can easily monitor, configure or update resources.
   Example: Cloud management platform, configuration tool [**16].**

9. Difference Between Structure and Behavior Documentation

Structure Documentation: Structure documentation describes the static, physical or logical structure of a system or network, showing the various components of the system, their interrelationships and their arrangement. It states “what is there” and “how it is arranged”. This type of documentation is static and does not describe the operational or dynamic aspects of the system [**12], [13].**

Examples:

- **In the case of computer systems:**
  - Hardware components of the system such as CPU, RAM, hard drive, motherboard and their connections.
  - Software architecture such as client-server model, microservices.
- **In the case of networks:**
  - Network topology such as star, ring or mesh topology.
  - Network devices such as routers, switches, firewalls and their physical connections.
  - IP address plan, VLAN configuration and subnet masking.

Behavioral Documentation: Behavioral documentation describes the dynamic, functional, or operational aspects of a system or network, answering the questions "how it works" and "how it responds". It describes how the various components of the system interact with each other, how information flows, and how the system responds to a particular input. It is like a manual, showing how specific tasks are accomplished **[14], [15].**

Examples:

- **In the case of computer systems:**
  - The input-output processing of the system.
  - Description of user interactions or user stories.
  - The operation of a function or algorithm.
  - How data flows from one component to another.
- **In the case of networks:**
  - How network protocols such as TCP/IP or DHCP work.
  - How data packets are transferred from one router to another.
  - The failover process of the network and how the system responds to an error.

Difference

Features	Structure Documentation	Behavior Documentation
Main Focus	Fixed structure or work or arrangement	Dynamic process
Examples	Network topology, hardware layout	Data packet flow, protocol operation
Nature	Blueprint or map	Guide

Main Functions:

- This type of file system primarily stores and distributes files.
- It creates and controls replication, which properly locates, stores, and removes files at specific locations.
- It also provides a unique user experience **[12], [13], [14]**.

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